Flutter Basics & Dart Language

1. What is Flutter?

Flutter is an open-source UI framework by Google. It allows building cross-platform apps (Android, iOS, Web, Desktop) with one codebase. Written in Dart language.

2. Why use Flutter?

• Cross-platform (Android, iOS, Web, Desktop)
• Hot Reload for fast development
• Near-native performance
• Beautiful Material & Cupertino widgets
• Strong Google & community support

3. Why is Flutter Important?

• Saves time & cost (single codebase)
• Faster time to market
• Fewer developers needed
• Used by big companies (Google Ads, BMW, Alibaba)
• Supports mobile, web & desktop

4. Flutter vs Other Frameworks

Flutter uses its own rendering engine (Skia), unlike React Native or Xamarin which rely on native components. This gives consistent UI & better performance.

5. Basic Flutter Concepts

• Everything is a Widget (Stateless, Stateful)
• Uses Dart language
• Hot Reload
• Widget Tree (Column, Row, Container, Stack)
• Material & Cupertino Widgets
• Navigation with Navigator
• State Management (setState, Provider, Bloc)
• Packages & Plugins (pub.dev)
• Cross-platform Support
• Rendering Engine (Skia)

6. Dart Language

Dart is an object-oriented language by Google, used for Flutter. It supports async programming, null safety, and compiles to native code.

Benefits of Dart:

• Cross-platform
• Fast performance
• Easy syntax
• Null Safety
• Async support

7. Dart Class & Object

Class = Blueprint, Object = Instance of class.
Example:

```
class Car {
  String brand = "";
  int year = 0;
  void display() {
    print("Car: $brand, Year: $year");
  }
}
```

8. Dart Variables & Data Types

• int, double, String, bool
• var, dynamic
• List (arrays)
• Map (key-value pairs)

9. Basic Syntax of Dart

Every program starts with main():
void main() { print('Hello Dart!'); }
Includes: variables, constants, operators, loops, functions, classes, collections.

10. Stateless vs Stateful Widget

• StatelessWidget → Does not change (static UI)
• StatefulWidget → Can change (dynamic UI)

Example:
Stateless: Label/Text
Stateful: Counter App with + button